Hyunchul Park

Ph.D. Student
Cho Chun Shik Graduate School of Mobility
Korea Advanced Institute of Science and Technology (KAIST)

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Research Interests

Humanoid Robot Drivers, Human Factors and Human–Machine Interaction in Traffic Safety, Driving Simulator and VR Experiments, Safety Testing of Automated Vehicles, Reinforcement Learning, V2X

Education

Ph.D. Student 2025 – *Present*
Cho Chun Shik Graduate School of Mobility, KAIST
Research: closing the embodiment gap for humanoid robot drivers
Advisor: Inhi Kim, Ph.D.

M.S. 2023 – 2025
Cho Chun Shik Graduate School of Mobility, KAIST
Research: driver human factors in driving-simulator and VR experiments
Advisor: Inhi Kim, Ph.D.

B.S. in Civil & Traffic Engineering 2016 – 2022
Kongju National University

Publications

Articles in International Journals

1. Baek, J., Kim, J., ***Park, H.***, & Kim, I. (2026). How age-related constraints reshape transport disadvantage in dynamic railway information systems: Evidence from virtual aging simulation. *Journal of Urban Mobility*, 10, 100292. https://doi.org/10.1016/j.urbmob.2026.100292
2. ***Park, H.***, Oh, T., Lim, J., & Kim, I. (2026). Micro mobility safety challenges: A study on drivers overtaking bicycles and e-scooters according to road conditions and prior riding experience. *Transportation Research Part F: Traffic Psychology and Behaviour*, 117, 103482. https://doi.org/10.1016/j.trf.2025.103482
3. Lim, J., ***Park, H.***, Oh, T., & Kim, I. (2025). The impact of information delivery systems in tunnels depending on lighting intensity and speed limit. *Tunnelling and Underground Space Technology*, 157, 106365. https://doi.org/10.1016/j.tust.2025.106365

4. ***Park, H.***, Oh, T., & Kim, I. (2024). Effects of driver's braking behavior by the real-time pedestrian scale warning system. *Accident Analysis & Prevention*, 205, 107685. https://doi.org/10.1016/j.aap.2024.107685

Articles in Domestic Journals (in Korean)

1. ***Park, H.***, Oh, T., & Kim, I. (2022). A study on validation of omnidirectional VR treadmill by comparison of spatial orientation skills. *The Journal of The Korea Institute of Intelligent Transport Systems*, 21(5), 15–27.

Articles Submitted

1. ***Park, H.***, Oh, T., Kim, J., Kim, J., & Kim, I. (2026). From passive visibility to active permissibility: A virtual reality study on overcoming track avoidance bias in subway station fires. *Underground Space (under 1st revision)*.
2. ***Park, H.***, Oh, T., Liu, Y., & Kim, I. (2026). Fair adversarial testing of automated vehicles in highway cut-ins: Reference-shielded attacks and a capability-controlled evaluation. *Transportation Research Part C (under review)*.
3. Kim, J., Kim, M., Kim, J., ***Park, H.***, & Kim, I. (2026). Risks of static obstacle placement in crowd evacuation: Density sensitivity and the isolation of beneficial zones. *(under review)*.

Manuscripts in Preparation

1. ***Park, H.***, & Kim, I. When the car moves the driver: Body loops and the speed limit of a simulated humanoid driver.
2. ***Park, H.***, & Kim, I. Seeing from a swaying body: Designing head-camera compensation for humanoid robot drivers from off-vehicle tests.
3. ***Park, H.***, Tamakloe, R., Kim, E. M., Lim, J., & Kim, I. Human-centric evaluation of UWB-based V2P ADAS: How pedestrian tag adoption rates influence driver cognitive load and safety performance in school zones.

Conferences

Articles in International Conference Proceedings

1. ***Park, H.***, & Kim, I. (2026, July). Severity-controllable adversarial testing of highway automated vehicles using naturalistic aggressive-moment statistics. In *15th Asia-Pacific Conference on Transportation and the Environment (APTE)*, Jeju, Korea. Oral presentation.
2. ***Park, H.***, Oh, T., Lim, J., & Kim, I. (2025). Micromobility safety challenges: A study on drivers overtaking bicycles and e-scooters according to road conditions and cross-modal experience. In *Transportation Research Board 104th Annual Meeting*, Washington, DC. Oral presentation; **TRB Outstanding Paper Award**.
3. ***Park, H.***, Oh, T., Lim, J., & Kim, I. (2024). Micro mobility safety challenges: A study on drivers overtaking bicycles and e-scooters in relation to road conditions and prior riding experience. In *36th ICTCT Conference*, The Hague, Netherlands. Oral presentation.
4. Lim, J., ***Park, H.***, Oh, T., & Kim, I. (2024). The impact of information delivery systems in tunnels depending on lighting intensity and speed limit. In *Transportation Research Board 103rd Annual Meeting*, Washington, DC. Poster.

5. ***Park, H.***, Oh, T., & Kim, I. (2023). Effects of driver's braking response by the real-time pedestrian scale warning system. In *Transportation Research Board 102nd Annual Meeting*, Washington, DC. Poster.
6. ***Park, H.***, Oh, T., & Kim, I. (2022). Validation of spatial orientation skill using virtual reality treadmill. In *ITS World Congress 2022*, Los Angeles, CA. Oral presentation.

Research Grants and Projects

Current Projects

1. **World-Model-Based Driving Skill Learning for Humanoid Robots and Real-Vehicle Validation in a V2X Testbed** 2026 – 2028
   National Research Foundation of Korea (NRF), Doctoral Student Research Encouragement Program
   Role: Principal Investigator, Amount: KRW 50,000,000
2. **Development of Industry-Specific Physical AI Foundation Models** 2026 – 2029
   Institute of Information & Communications Technology Planning & Evaluation (IITP)
   Role: Researcher (sim-to-real module for a logistics humanoid)
3. **Development of Virtual Environment and Demonstration Technology for Automated Driving based on Metaverse** 2023 – 2027
   Ministry of Land, Infrastructure and Transport (MOLIT)
   Role: Researcher (behaviorally realistic traffic agents for automated-driving tests)

Selected Completed Projects

1. **Developing an Aggressive Pedestrian Model for Autonomous Vehicle Safety Testing** 2024 – 2025
   National Research Foundation of Korea (NRF), Graduate Student Research Encouragement Fund
   Role: Principal Investigator, Amount: KRW 12,000,000
2. **ASEAN–Korea Transport Cooperation Roadmap (2026–2030)** 2024 – 2025
   Ministry of Land, Infrastructure and Transport (MOLIT), with the ASEAN Secretariat
   Role: Planning lead; presented at the ASEAN Land Transport Working Group, Cambodia, Aug. 2025

Awards and Honors

2026 National Excellence Scholarship (Science & Engineering), *Korea Student Aid Foundation*
2025 Outstanding Paper Award, *Transportation Research Board 104th Annual Meeting*
2025 Young Investigator Award, *IPESK*
2024 Graduate Student Outstanding Paper Award, *KAIST*
2024 Young-Han Kim Global Leader Scholarship, *KAIST* (25 students annually)
2024 Best Teaching Assistant Award, *KAIST*
2024 President's Award (Distinguished Graduate), *Korean Society of Transportation*
2023 Best Paper Award, *The Korea Institute of Intelligent Transport Systems*

2021 Best Paper Award, *The Korea Institute of Intelligent Transport Systems*
2021 Excellence Award, Marine Big Data Analysis Competition, *UNIST*
2020 President's Award (Capstone Design), *Kongju National University*

Talks

1. "Building an LLM-Powered Wiki for Researchers: Toward Automated Knowledge Curation in Research Workflows," AI Agent Seminar, Korea Transport Institute (KOTI), Jun. 2026.

Teaching

Teaching Assistant 2024
Experiments in Future Mobility within the Metaverse, KAIST
Best Teaching Assistant Award

Last updated: October 2026